

Project_4_Solution

```
[1]: # loading packages
import os
import pandas as pd
import numpy as np

# plotting packages
%matplotlib inline
from mpl_toolkits.mplot3d import Axes3D
import matplotlib.pyplot as plt
import matplotlib.cm as cm
import matplotlib.colors as clr

# Kmeans algorithm from scikit-learn
from sklearn.cluster import KMeans
```

0.1 Load raw data

```
[2]: # load raw data
raw = pd.read_csv('countryriskdata.csv')

# check the raw data
print("Size of the dataset (row, col): ", raw.shape)
print("\nFirst 5 rows\n", raw.head(n=5))
```

Size of the dataset (row, col): (122, 6)

First 5 rows

	Country	Abbrev	Corruption	Peace	Legal	GDP Growth
0	Albania	AL	39	1.867	3.822	3.403
1	Algeria	DZ	34	2.213	4.160	4.202
2	Argentina	AR	36	1.957	4.568	-2.298
3	Armenia	AM	33	2.218	4.126	0.208
4	Australia	AU	79	1.465	8.244	2.471

0.2 Simple exploratory analysis

0.2.1 Print summary statistics

Note that all features have quite different variances, and Corruption and Legal are highly correlated.

```
[3]: # print summary statistics
print("\nSummary statistics\n", raw.describe().round(3))
# Plot the correlation matrix
print("\nCorrelation matrix\n")
corr = raw.corr()
corr.style.background_gradient(cmap='coolwarm').set_precision(3)
```

Summary statistics

	Corruption	Peace	Legal	GDP Growth
count	122.000	122.000	122.000	122.000
mean	46.238	2.004	5.599	2.373
std	19.126	0.448	1.487	3.241
min	14.000	1.192	2.728	-18.000
25%	31.250	1.685	4.572	1.432
50%	40.000	1.969	5.274	2.496
75%	58.750	2.280	6.477	4.080
max	90.000	3.399	8.633	7.958

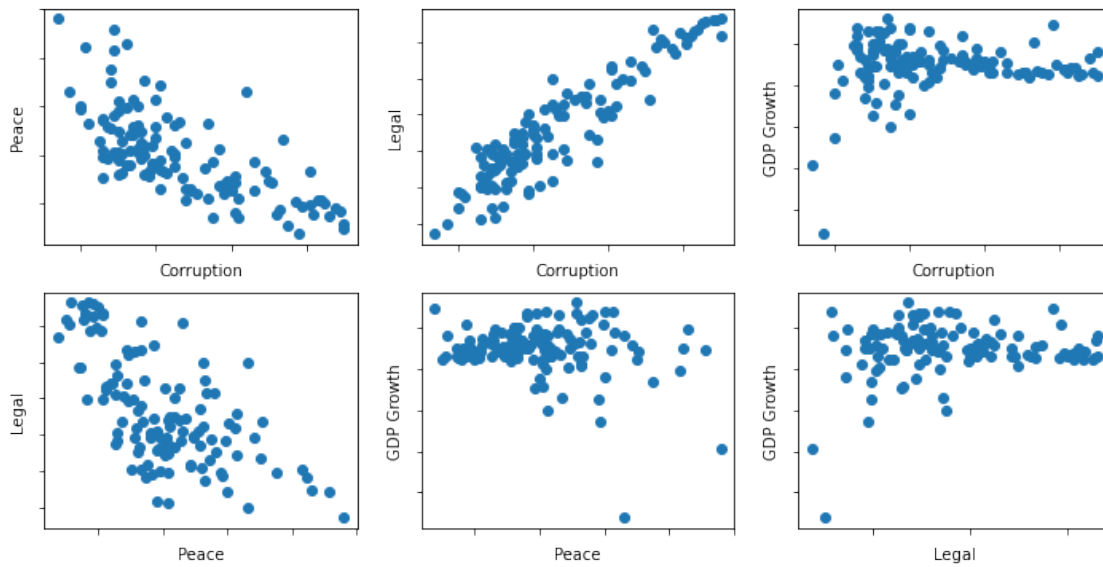
Correlation matrix

```
[3]: <pandas.io.formats.style.Styler at 0x1caf184da30>
```

```
[4]: # Scatter plots for different pairs of variables
fig, axs = plt.subplots(2, 3, figsize=(12,6))
fig.suptitle('Scatter plot among each pair of the variables')
Labels = ['Corruption', 'Peace', 'Legal', 'GDP Growth']

# Plot the Figure
m = 0
for i in range(len(Labels)-1):
    for j in range(1, len(Labels)-i):
        axs.flat[m].scatter(raw[Labels[i]], raw[Labels[i+j]])
        axs.flat[m].set_xlabel(Labels[i])
        axs.flat[m].set_ylabel(Labels[i+j])
        # Turn off tick labels.
        axs.flat[m].set_yticklabels([])
        axs.flat[m].set_xticklabels([])
    m = m+1
```

Scatter plot among each pair of the variables



0.2.2 Plot histogram

Note that the distribution for GDP Growth is quite skewed.

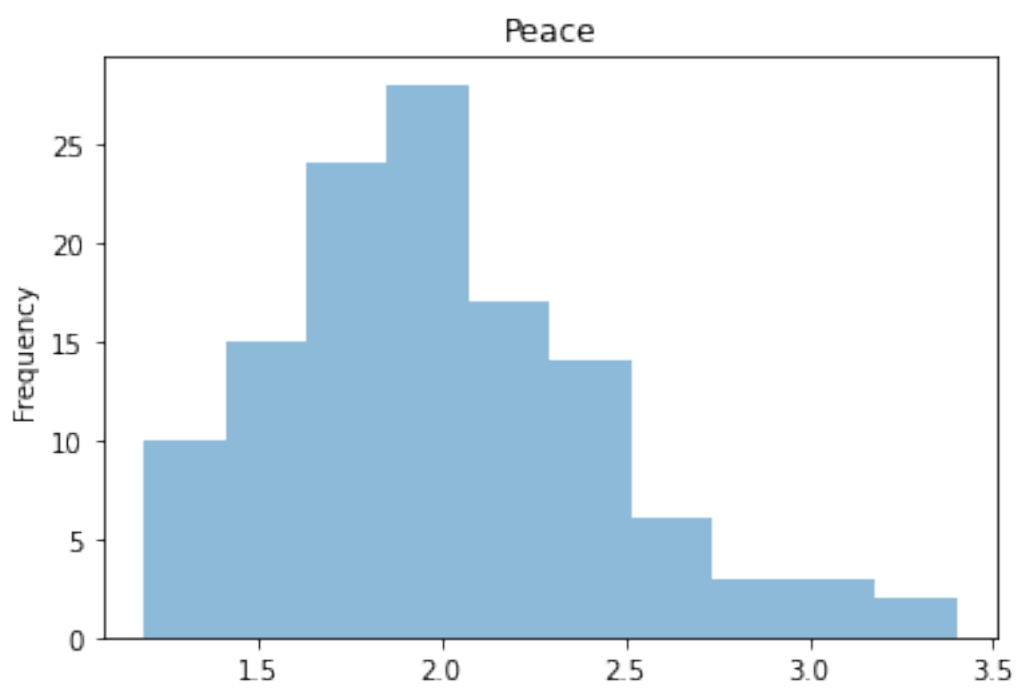
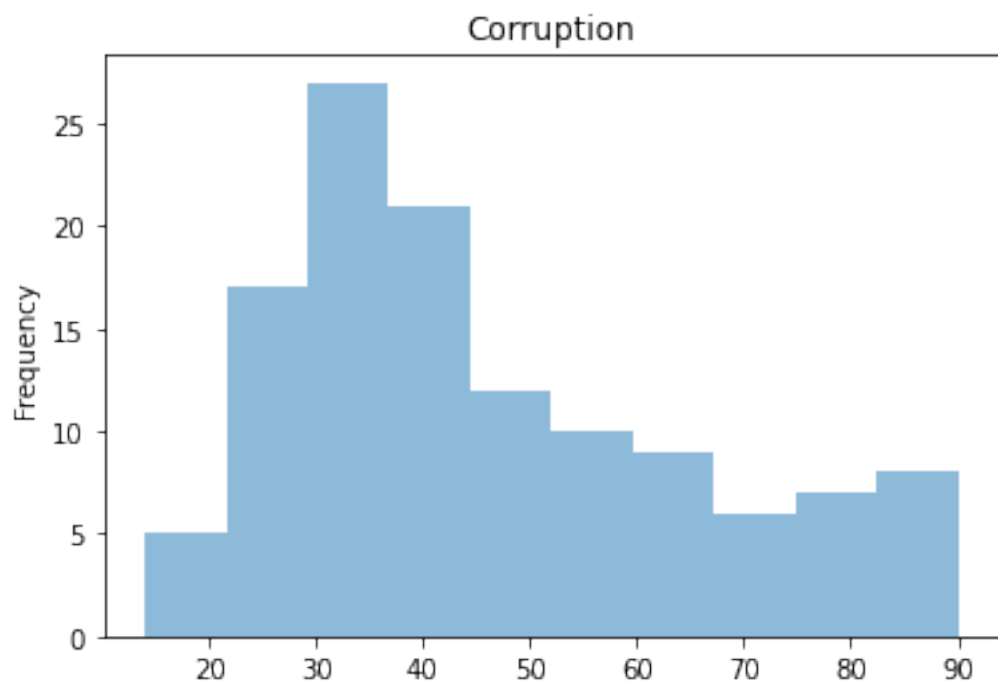
```
[5]: # plot histograms
plt.figure(1)
raw['Corruption'].plot(kind = 'hist', title = 'Corruption', alpha = 0.5)

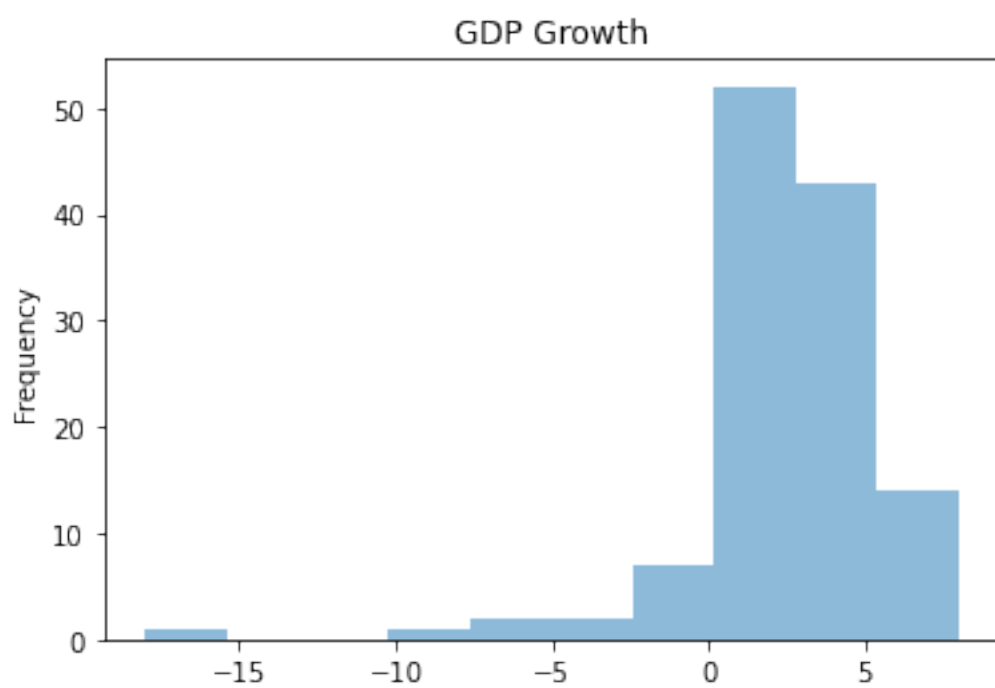
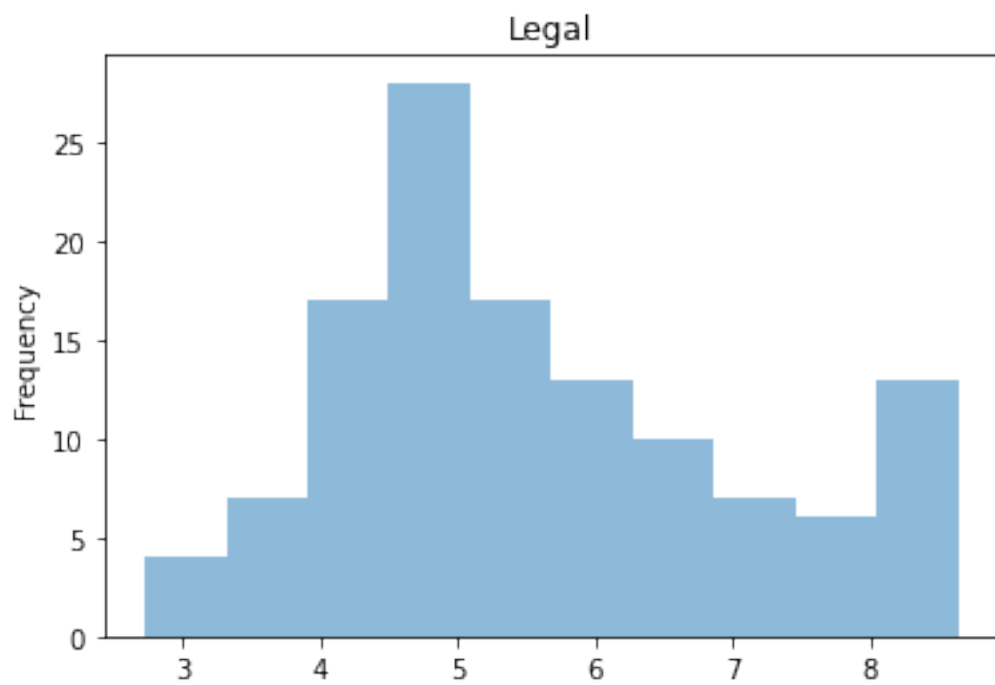
plt.figure(2)
raw['Peace'].plot(kind = 'hist', title = 'Peace', alpha = 0.5)

plt.figure(3)
raw['Legal'].plot(kind = 'hist', title = 'Legal', alpha = 0.5)

plt.figure(4)
raw['GDP Growth'].plot(kind = 'hist', title = 'GDP Growth', alpha = 0.5)

plt.show()
```





0.3 K-means cluster

0.3.1 Pick features and normalization

Let's normalize all the features, effectively making them equally weighted.

Ref. [Feature normalization](#).

```
[6]: X = raw[['Corruption', 'Peace', 'Legal', 'GDP Growth']]
X = (X - X.mean()) / X.std()
print(X.head(5))
```

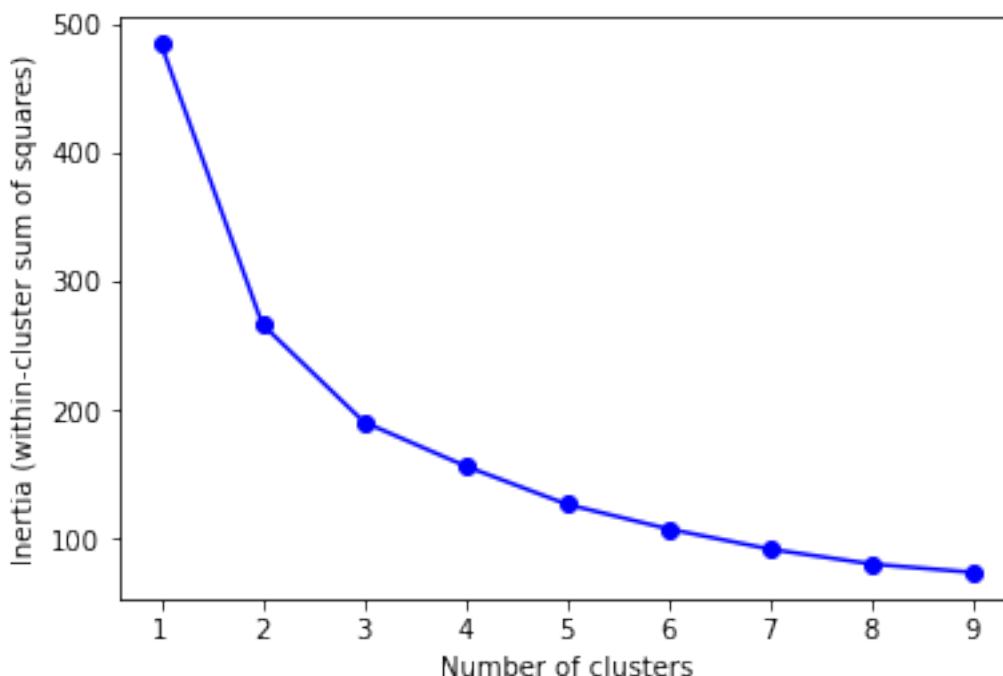
	Corruption	Peace	Legal	GDP Growth
0	-0.378414	-0.305319	-1.194666	0.317896
1	-0.639833	0.467304	-0.967413	0.564392
2	-0.535266	-0.104348	-0.693096	-1.440899
3	-0.692117	0.478469	-0.990273	-0.667782
4	1.712936	-1.202990	1.778450	0.030368

0.3.2 Perform elbow method

The marginal gain of adding one cluster dropped quite a bit from k=3 to k=4. We will choose k=3 (not a clear cut though).

Ref. [Determining the number of clusters in a dataset](#).

```
[7]: # https://stackoverflow.com/questions/41540751/sklearn-kmeans-equivalent-of-elbow-method
Ks = range(1,10)
inertia = [KMeans(i).fit(X).inertia_ for i in Ks]
fig = plt.figure()
plt.plot(Ks, inertia, '-bo')
plt.xlabel('Number of clusters')
plt.ylabel('Inertia (within-cluster sum of squares)')
plt.show()
```



0.3.3 K-means with $k = 3$

```
[8]: k = 3
kmeans = KMeans(n_clusters=k,random_state=0)
kmeans.fit(X)

# print inertia & cluster center
print("inertia for k=3 is", np.round(kmeans.inertia_,3))

Cluster_Table = pd.DataFrame(data=kmeans.cluster_centers_,
    ↳columns=['Corruption', 'Peace', 'Legal', 'GDP Growth'])
Cluster_Table['Cluster'] = Cluster_Table.index
print("\ncluster centers:\n")
print(Cluster_Table.to_string(index=False))

# take a quick look at the result for each country
raw['Cluster Lables']=kmeans.labels_
pd.set_option("display.max_rows", None) # Show the table in a scrollable window
raw
```

inertia for k=3 is 189.68

cluster centers:

Corruption	Peace	Legal	GDP Growth	Cluster
------------	-------	-------	------------	---------

-1.023248	1.346664	-1.054684	-1.712920	0
-0.462523	0.247951	-0.429546	0.364722	1
1.243758	-0.981804	1.196287	0.013895	2

[8]:

	Country	Abbrev	Corruption	Peace	Legal	\
0	Albania	AL	39	1.867	3.822	
1	Algeria	DZ	34	2.213	4.160	
2	Argentina	AR	36	1.957	4.568	
3	Armenia	AM	33	2.218	4.126	
4	Australia	AU	79	1.465	8.244	
5	Austria	AT	75	1.278	8.012	
6	Azerbaijan	AZ	30	2.450	3.946	
7	Bahrain	BH	43	2.398	6.157	
8	Bangladesh	BD	26	2.045	3.117	
9	Belgium	BE	77	1.528	7.839	
10	Benin	BJ	36	1.998	4.583	
11	Bolivia	BO	33	2.038	3.971	
12	Bosnia and Herzegovina	BA	39	1.915	3.917	
13	Botswana	BW	60	1.639	6.126	
14	Brazil	BR	40	2.176	5.434	
15	Bulgaria	BG	41	1.646	4.813	
16	Burundi	BI	20	2.500	3.430	
17	Cameroon	CM	26	2.356	4.294	
18	Canada	CA	82	1.388	8.179	
19	Chad	TD	20	2.464	3.891	
20	Chile	CL	66	1.635	6.926	
21	China	CN	40	2.288	5.712	
22	Colombia	CO	37	2.764	5.354	
23	Costa Rica	CR	58	1.699	6.060	
24	Croatia	HR	49	1.633	4.754	
25	Cyprus	CY	55	1.924	5.447	
26	Czech Republic	CI	55	1.360	6.860	
27	Democratic Republic of Congo	CI	21	3.112	3.818	
28	Denmark	DK	90	1.246	8.158	
29	Dominican Republic	DO	31	2.143	4.823	
30	Ecuador	EC	31	2.020	4.627	
31	Egypt	EG	34	2.574	4.433	
32	El Salvador	SV	36	2.237	4.945	
33	Estonia	EE	70	1.732	7.199	
34	Ethiopia	ET	34	2.284	4.718	
35	Finland	FI	89	1.429	8.626	
36	France	FR	69	1.829	7.336	
37	Gabon	GA	35	2.033	4.694	
38	Georgia	GE	57	2.057	4.707	
39	Germany	DE	81	1.486	7.959	
40	Ghana	GH	43	1.809	5.646	
41	Greece	GR	44	2.044	5.389	

42	Guatemala	GT	28	2.270	5.077
43	Honduras	HN	30	2.237	4.902
44	Hungary	HU	48	1.534	5.992
45	Iceland	IS	78	1.192	7.700
46	India	IN	40	2.566	5.564
47	Indonesia	ID	37	1.799	5.167
48	Iran	ID	29	2.411	4.521
49	Ireland	IE	73	1.433	7.872
50	Israel	IL	64	2.656	6.974
51	Italy	IT	47	1.774	5.979
52	Jamaica	JM	39	2.091	6.010
53	Japan	JP	72	1.395	8.327
54	Jordan	JO	48	2.127	6.266
55	Kazakhstan	KZ	29	2.019	4.432
56	Kenya	KE	26	2.379	4.856
57	Korea (South)	KI	53	1.858	6.495
58	Kuwait	KW	41	1.842	5.380
59	Latvia	LV	57	1.680	5.341
60	Lebanon	LB	28	2.752	4.331
61	Liberia	LR	37	1.998	4.893
62	Lithuania	LT	59	1.735	5.918
63	Madagascar	MG	26	1.763	4.064
64	Malawi	MW	31	1.817	4.670
65	Malaysia	MY	49	1.648	6.610
66	Mali	ML	32	2.489	4.814
67	Mauritania	MR	27	2.295	4.091
68	Mauritius	MU	54	1.559	6.315
69	Mexico	MX	30	2.557	5.194
70	Moldova	FM	30	1.953	3.178
71	Montenegro	ME	45	1.884	4.192
72	Morocco	MA	37	2.086	5.500
73	Mozambique	MZ	27	1.963	4.625
74	Nepal	NP	29	2.026	4.941
75	Netherlands	NL	83	1.541	8.296
76	New Zealand	NZ	90	1.287	8.633
77	Nicaragua	NI	26	1.975	3.990
78	Nigeria	NG	28	2.877	3.950
79	Norway	NO	85	1.500	8.533
80	Oman	OM	45	2.016	6.280
81	Pakistan	PK	32	3.145	3.474
82	Panama	PA	38	1.837	5.798
83	Paraguay	PY	30	2.037	4.478
84	Peru	PE	35	2.057	5.217
85	Philippines	PH	35	2.511	5.331
86	Poland	PL	62	1.557	6.253
87	Portugal	PT	62	1.356	6.848
88	Qatar	QA	61	1.716	7.348

89	Romania	RO	48	1.649	5.042
90	Russia	RO	29	3.079	4.043
91	Rwanda	RW	54	2.323	6.508
92	Saudi Arabia	SA	46	2.338	6.133
93	Senegal	SN	45	1.978	4.942
94	Serbia	RS	42	1.834	4.044
95	Sierra Leone	SL	30	1.805	4.523
96	Singapore	SG	84	1.535	8.358
97	Slovakia	SK	51	1.603	6.396
98	Slovenia	SI	61	1.408	5.994
99	South Africa	ZA	45	2.316	7.000
100	Spain	ES	58	1.604	6.422
101	Sri Lanka	LK	36	2.133	5.390
102	Sweden	SE	88	1.461	8.608
103	Switzerland	CH	86	1.370	8.561
104	Taiwan	SY	61	1.787	7.268
105	Tanzania	TJ	32	1.899	5.051
106	Thailand	TJ	35	2.312	5.215
107	The FYR of Macedonia	TJ	37	2.092	4.655
108	Trinidad and Tobago	TT	35	2.056	5.503
109	Tunisia	TN	41	1.949	5.081
110	Turkey	TR	41	2.710	4.925
111	Uganda	UG	25	2.148	5.102
112	Ukraine	UA	29	3.287	3.424
113	United Arab Emirates	AE	66	1.931	7.483
114	United Kingdom	GB	81	1.830	8.129
115	United States	US	74	2.154	8.074
116	Uruguay	UY	71	1.726	6.412
117	Venezuela	VE	17	2.651	3.000
118	Vietnam	VI	33	1.906	4.930
119	Yemen	YE	14	3.399	2.728
120	Zambia	ZM	38	1.783	4.917
121	Zimbabwe	ZW	22	2.322	3.760

	GDP Growth	Cluster Lables
0	3.403	1
1	4.202	1
2	-2.298	0
3	0.208	1
4	2.471	2
5	1.482	2
6	-3.772	0
7	2.915	1
8	6.923	1
9	1.239	2
10	4.027	1
11	4.100	1

12	2.500	1
13	2.864	2
14	-3.595	0
15	3.443	1
16	-1.042	0
17	4.443	1
18	1.433	2
19	-6.373	0
20	1.555	2
21	6.700	1
22	1.960	1
23	4.329	2
24	2.930	1
25	2.836	1
26	2.408	2
27	2.400	0
28	1.137	2
29	6.567	1
30	-2.168	0
31	4.298	1
32	2.400	1
33	1.573	2
34	7.958	1
35	1.387	2
36	1.213	2
37	2.263	1
38	2.723	1
39	1.774	2
40	4.041	1
41	0.012	1
42	3.000	1
43	3.600	1
44	2.000	2
45	7.200	2
46	6.828	1
47	5.016	1
48	6.544	1
49	5.222	2
50	4.003	1
51	0.880	1
52	1.540	1
53	0.999	2
54	2.100	1
55	1.083	1
56	6.004	1
57	2.828	2
58	2.458	1

59	1.954	1
60	1.000	0
61	-1.170	1
62	2.274	2
63	4.138	1
64	2.270	1
65	4.239	2
66	5.370	1
67	1.522	1
68	3.600	2
69	2.302	1
70	4.000	1
71	2.358	1
72	1.487	1
73	3.400	1
74	0.561	1
75	2.095	2
76	3.956	2
77	4.700	1
78	-1.541	0
79	1.024	2
80	3.057	1
81	4.707	1
82	5.000	1
83	4.093	1
84	3.897	1
85	6.839	1
86	2.830	2
87	1.432	2
88	2.683	2
89	4.785	1
90	-0.248	0
91	5.932	1
92	1.398	1
93	6.574	1
94	2.779	1
95	4.903	1
96	1.996	2
97	3.285	2
98	2.492	2
99	0.279	1
100	3.235	2
101	4.295	1
102	3.310	2
103	1.310	2
104	1.402	2
105	6.579	1

106	3.230	1
107	2.406	1
108	-5.110	0
109	1.000	1
110	2.876	1
111	4.673	1
112	2.308	0
113	2.732	2
114	1.806	2
115	1.616	2
116	1.438	2
117	-18.000	0
118	6.211	1
119	-9.779	0
120	2.967	1
121	0.522	0

From the cluster centers, it appears that cluster 0 contains the countries with lowest risk, cluster 2 contains countries with medium risk and cluster 1 contains countries with highest risk

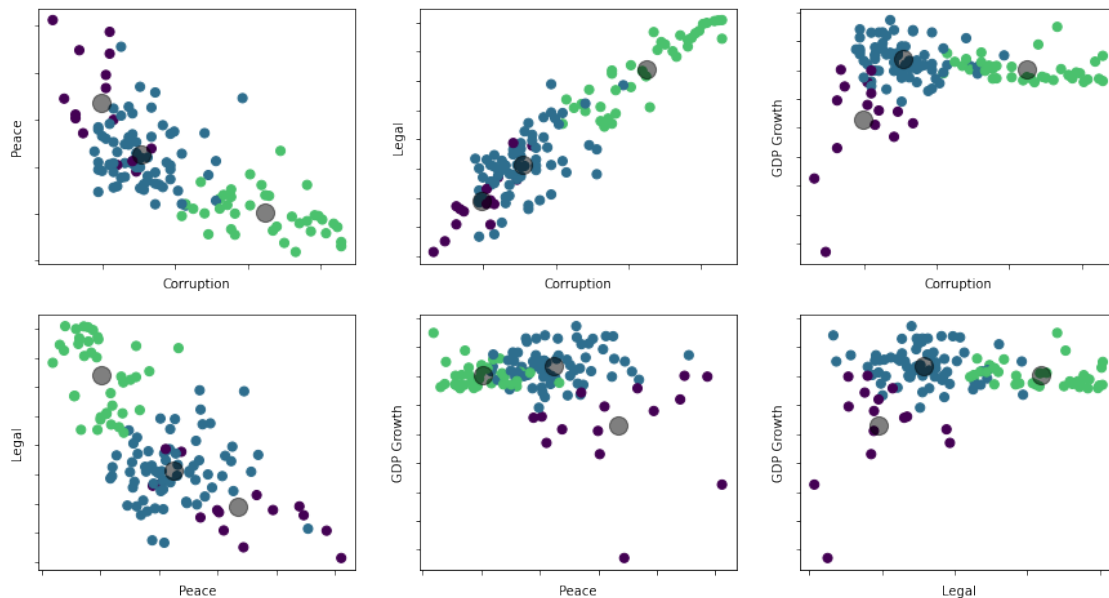
0.3.4 Visualize the result (2D plots)

```
[9]: # Scatter plots of different pairs of variables
fig, axs = plt.subplots(2, 3, figsize=(15,8))
fig.suptitle('Scatter plots for different pairs of variables')
Labels = ['Corruption', 'Peace', 'Legal', 'GDP Growth']
y = kmeans.labels_

# set up the color
norm = clr.Normalized(vmin=0., vmax=y.max()+0.8)
cmap = cm.viridis

# Plot the Figure
m = 0
for i in range(len(Labels)-1):
    for j in range(1, len(Labels)-i):
        axs.flat[m].scatter(X[Labels[i]], X[Labels[i+j]], c=cmap(norm(y)), s=50)
        axs.flat[m].scatter(Cluster_Table[Labels[i]],
        ↪Cluster_Table[Labels[i+j]], c='black', s=200, alpha=0.5)
        axs.flat[m].set_xlabel(Labels[i])
        axs.flat[m].set_ylabel(Labels[i+j])
        # Turn off tick labels.
        axs.flat[m].set_yticklabels([])
        axs.flat[m].set_xticklabels([])
    m = m+1
```

Scatter plots for different pairs of variables



0.3.5 Visualize the result (2D plots with names of countries)

plot country abbreviations instead of dots.

```
[10]: # Plot of the Scatter Figure among each pair
fig, axs = plt.subplots(2, 3, figsize=(15,8))
fig.suptitle('Scatter plot among each pair of the variables')
Labels=['Corruption','Peace','Legal','GDP Growth']
colors = ['blue','green', 'red']

# set up the color
norm = clr.Normalized(vmin=0.,vmax=y.max()+0.8)
cmap = cm.viridis

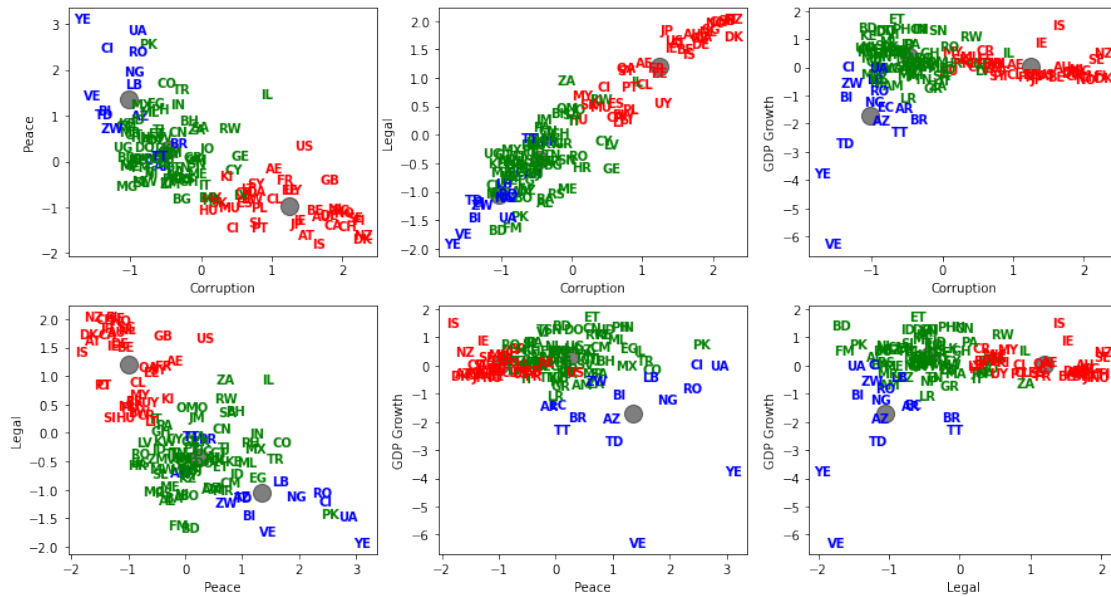
# Plot the Figure
m = 0
for i in range(len(Labels)-1):
    for j in range(1,len(Labels)-i):
        axs.flat[m].scatter(X[Labels[i]],X[Labels[i+j]],c=y, s=0, alpha=0)
        axs.flat[m].scatter(Cluster_Table[Labels[i]],
        ↪Cluster_Table[Labels[i+j]], c='black', s=200, alpha=0.5)
        axs.flat[m].set_xlabel(Labels[i])
        axs.flat[m].set_ylabel(Labels[i+j])
        for h in range(X.shape[0]):
```

```

        axs.flat[m].text(X[Labels[i]].iloc[h], X[Labels[i+j]].iloc[h],
        ↪raw['Abbrev'].iloc[h],
        color=colors[y[h]], weight='semibold', horizontalalignment =
        ↪'center', verticalalignment = 'center')
        m = m+1

```

Scatter plot among each pair of the variables



0.3.6 List the result

```

[11]: result = pd.DataFrame({'Country':raw['Country'], 'Abbrev':raw['Abbrev'],
        ↪'Label':y})
with pd.option_context('display.max_rows', None, 'display.max_columns', 3):
    print(result.sort_values(by=['Label', 'Country']))

```

	Country	Abbrev	Label
2	Argentina	AR	0
6	Azerbaijan	AZ	0
14	Brazil	BR	0
16	Burundi	BI	0
19	Chad	TD	0
27	Democratic Republic of Congo	CI	0
30	Ecuador	EC	0
60	Lebanon	LB	0
78	Nigeria	NG	0
90	Russia	RO	0
108	Trinidad and Tobago	TT	0
112	Ukraine	UA	0

117	Venezuela	VE	0
119	Yemen	YE	0
121	Zimbabwe	ZW	0
0	Albania	AL	1
1	Algeria	DZ	1
3	Armenia	AM	1
7	Bahrain	BH	1
8	Bangladesh	BD	1
10	Benin	BJ	1
11	Bolivia	BO	1
12	Bosnia and Herzegovina	BA	1
15	Bulgaria	BG	1
17	Cameroon	CM	1
21	China	CN	1
22	Colombia	CO	1
24	Croatia	HR	1
25	Cyprus	CY	1
29	Dominican Republic	DO	1
31	Egypt	EG	1
32	El Salvador	SV	1
34	Ethiopia	ET	1
37	Gabon	GA	1
38	Georgia	GE	1
40	Ghana	GH	1
41	Greece	GR	1
42	Guatemala	GT	1
43	Honduras	HN	1
46	India	IN	1
47	Indonesia	ID	1
48	Iran	ID	1
50	Israel	IL	1
51	Italy	IT	1
52	Jamaica	JM	1
54	Jordan	JO	1
55	Kazakhstan	KZ	1
56	Kenya	KE	1
58	Kuwait	KW	1
59	Latvia	LV	1
61	Liberia	LR	1
63	Madagascar	MG	1
64	Malawi	MW	1
66	Mali	ML	1
67	Mauritania	MR	1
69	Mexico	MX	1
70	Moldova	FM	1
71	Montenegro	ME	1
72	Morocco	MA	1
73	Mozambique	MZ	1

74	Nepal	NP	1
77	Nicaragua	NI	1
80	Oman	OM	1
81	Pakistan	PK	1
82	Panama	PA	1
83	Paraguay	PY	1
84	Peru	PE	1
85	Philippines	PH	1
89	Romania	RO	1
91	Rwanda	RW	1
92	Saudi Arabia	SA	1
93	Senegal	SN	1
94	Serbia	RS	1
95	Sierra Leone	SL	1
99	South Africa	ZA	1
101	Sri Lanka	LK	1
105	Tanzania	TJ	1
106	Thailand	TJ	1
107	The FYR of Macedonia	TJ	1
109	Tunisia	TN	1
110	Turkey	TR	1
111	Uganda	UG	1
118	Vietnam	VI	1
120	Zambia	ZM	1
4	Australia	AU	2
5	Austria	AT	2
9	Belgium	BE	2
13	Botswana	BW	2
18	Canada	CA	2
20	Chile	CL	2
23	Costa Rica	CR	2
26	Czech Republic	CI	2
28	Denmark	DK	2
33	Estonia	EE	2
35	Finland	FI	2
36	France	FR	2
39	Germany	DE	2
44	Hungary	HU	2
45	Iceland	IS	2
49	Ireland	IE	2
53	Japan	JP	2
57	Korea (South)	KI	2
62	Lithuania	LT	2
65	Malaysia	MY	2
68	Mauritius	MU	2
75	Netherlands	NL	2
76	New Zealand	NZ	2
79	Norway	NO	2

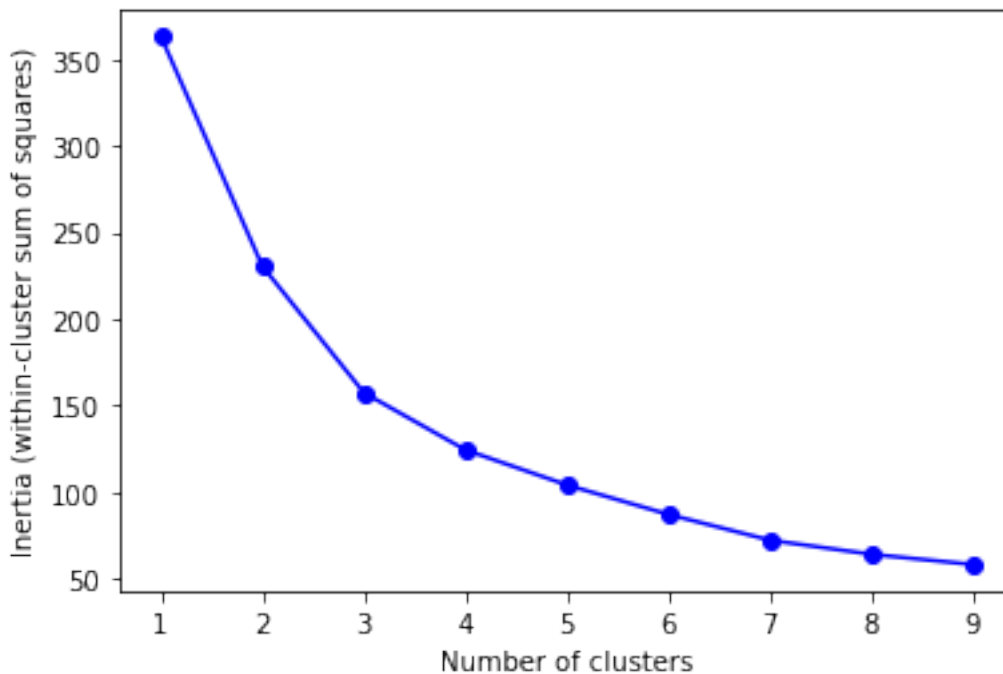
86	Poland	PL	2
87	Portugal	PT	2
88	Qatar	QA	2
96	Singapore	SG	2
97	Slovakia	SK	2
98	Slovenia	SI	2
100	Spain	ES	2
102	Sweden	SE	2
103	Switzerland	CH	2
104	Taiwan	SY	2
113	United Arab Emirates	AE	2
114	United Kingdom	GB	2
115	United States	US	2
116	Uruguay	UY	2

0.3.7 Since Corruption and Legal are highly correlated, we drop the Corruption variable, i.e., we pick three features for this analysis, Peace, Legal and GDP Grwoth.

```
[12]: X = X[['Peace', 'Legal', 'GDP Growth']]
```

```
[13]: Ks = range(1, 10)
inertia = [KMeans(i).fit(X).inertia_ for i in Ks]

fig = plt.figure()
plt.plot(Ks, inertia, '-bo')
plt.xlabel('Number of clusters')
plt.ylabel('Inertia (within-cluster sum of squares)')
plt.show()
```



0.3.8 K-means with $k = 3$

```
[14]: k = 3
kmeans = KMeans(n_clusters=k,random_state=0)
kmeans.fit(X)

# print inertia & cluster center
print("inertia for k=3 is", np.round(kmeans.inertia_,3))

Cluster_Table = pd.DataFrame(data=kmeans.cluster_centers_, columns=['Peace', 'Legal', 'GDP Growth'])
Cluster_Table['Cluster'] = Cluster_Table.index
print("\ncluster centers:\n")
print(Cluster_Table.to_string(index=False))

# take a quick look at the result for each country
raw['Cluster Lables']=kmeans.labels_
pd.set_option("display.max_rows", None) # Show the table in a scrollable window
raw
```

inertia for k=3 is 157.551

cluster centers:

Peace	Legal	GDP Growth	Cluster
-------	-------	------------	---------

-0.928106	1.166413	-0.014458	0
1.215626	-1.016771	-1.614970	1
0.253209	-0.451868	0.431274	2

[14]:

	Country	Abbrev	Corruption	Peace	Legal	\
0	Albania	AL	39	1.867	3.822	
1	Algeria	DZ	34	2.213	4.160	
2	Argentina	AR	36	1.957	4.568	
3	Armenia	AM	33	2.218	4.126	
4	Australia	AU	79	1.465	8.244	
5	Austria	AT	75	1.278	8.012	
6	Azerbaijan	AZ	30	2.450	3.946	
7	Bahrain	BH	43	2.398	6.157	
8	Bangladesh	BD	26	2.045	3.117	
9	Belgium	BE	77	1.528	7.839	
10	Benin	BJ	36	1.998	4.583	
11	Bolivia	BO	33	2.038	3.971	
12	Bosnia and Herzegovina	BA	39	1.915	3.917	
13	Botswana	BW	60	1.639	6.126	
14	Brazil	BR	40	2.176	5.434	
15	Bulgaria	BG	41	1.646	4.813	
16	Burundi	BI	20	2.500	3.430	
17	Cameroon	CM	26	2.356	4.294	
18	Canada	CA	82	1.388	8.179	
19	Chad	TD	20	2.464	3.891	
20	Chile	CL	66	1.635	6.926	
21	China	CN	40	2.288	5.712	
22	Colombia	CO	37	2.764	5.354	
23	Costa Rica	CR	58	1.699	6.060	
24	Croatia	HR	49	1.633	4.754	
25	Cyprus	CY	55	1.924	5.447	
26	Czech Republic	CI	55	1.360	6.860	
27	Democratic Republic of Congo	CI	21	3.112	3.818	
28	Denmark	DK	90	1.246	8.158	
29	Dominican Republic	DO	31	2.143	4.823	
30	Ecuador	EC	31	2.020	4.627	
31	Egypt	EG	34	2.574	4.433	
32	El Salvador	SV	36	2.237	4.945	
33	Estonia	EE	70	1.732	7.199	
34	Ethiopia	ET	34	2.284	4.718	
35	Finland	FI	89	1.429	8.626	
36	France	FR	69	1.829	7.336	
37	Gabon	GA	35	2.033	4.694	
38	Georgia	GE	57	2.057	4.707	
39	Germany	DE	81	1.486	7.959	
40	Ghana	GH	43	1.809	5.646	
41	Greece	GR	44	2.044	5.389	

42	Guatemala	GT	28	2.270	5.077
43	Honduras	HN	30	2.237	4.902
44	Hungary	HU	48	1.534	5.992
45	Iceland	IS	78	1.192	7.700
46	India	IN	40	2.566	5.564
47	Indonesia	ID	37	1.799	5.167
48	Iran	ID	29	2.411	4.521
49	Ireland	IE	73	1.433	7.872
50	Israel	IL	64	2.656	6.974
51	Italy	IT	47	1.774	5.979
52	Jamaica	JM	39	2.091	6.010
53	Japan	JP	72	1.395	8.327
54	Jordan	JO	48	2.127	6.266
55	Kazakhstan	KZ	29	2.019	4.432
56	Kenya	KE	26	2.379	4.856
57	Korea (South)	KI	53	1.858	6.495
58	Kuwait	KW	41	1.842	5.380
59	Latvia	LV	57	1.680	5.341
60	Lebanon	LB	28	2.752	4.331
61	Liberia	LR	37	1.998	4.893
62	Lithuania	LT	59	1.735	5.918
63	Madagascar	MG	26	1.763	4.064
64	Malawi	MW	31	1.817	4.670
65	Malaysia	MY	49	1.648	6.610
66	Mali	ML	32	2.489	4.814
67	Mauritania	MR	27	2.295	4.091
68	Mauritius	MU	54	1.559	6.315
69	Mexico	MX	30	2.557	5.194
70	Moldova	FM	30	1.953	3.178
71	Montenegro	ME	45	1.884	4.192
72	Morocco	MA	37	2.086	5.500
73	Mozambique	MZ	27	1.963	4.625
74	Nepal	NP	29	2.026	4.941
75	Netherlands	NL	83	1.541	8.296
76	New Zealand	NZ	90	1.287	8.633
77	Nicaragua	NI	26	1.975	3.990
78	Nigeria	NG	28	2.877	3.950
79	Norway	NO	85	1.500	8.533
80	Oman	OM	45	2.016	6.280
81	Pakistan	PK	32	3.145	3.474
82	Panama	PA	38	1.837	5.798
83	Paraguay	PY	30	2.037	4.478
84	Peru	PE	35	2.057	5.217
85	Philippines	PH	35	2.511	5.331
86	Poland	PL	62	1.557	6.253
87	Portugal	PT	62	1.356	6.848
88	Qatar	QA	61	1.716	7.348

89	Romania	RO	48	1.649	5.042
90	Russia	RO	29	3.079	4.043
91	Rwanda	RW	54	2.323	6.508
92	Saudi Arabia	SA	46	2.338	6.133
93	Senegal	SN	45	1.978	4.942
94	Serbia	RS	42	1.834	4.044
95	Sierra Leone	SL	30	1.805	4.523
96	Singapore	SG	84	1.535	8.358
97	Slovakia	SK	51	1.603	6.396
98	Slovenia	SI	61	1.408	5.994
99	South Africa	ZA	45	2.316	7.000
100	Spain	ES	58	1.604	6.422
101	Sri Lanka	LK	36	2.133	5.390
102	Sweden	SE	88	1.461	8.608
103	Switzerland	CH	86	1.370	8.561
104	Taiwan	SY	61	1.787	7.268
105	Tanzania	TJ	32	1.899	5.051
106	Thailand	TJ	35	2.312	5.215
107	The FYR of Macedonia	TJ	37	2.092	4.655
108	Trinidad and Tobago	TT	35	2.056	5.503
109	Tunisia	TN	41	1.949	5.081
110	Turkey	TR	41	2.710	4.925
111	Uganda	UG	25	2.148	5.102
112	Ukraine	UA	29	3.287	3.424
113	United Arab Emirates	AE	66	1.931	7.483
114	United Kingdom	GB	81	1.830	8.129
115	United States	US	74	2.154	8.074
116	Uruguay	UY	71	1.726	6.412
117	Venezuela	VE	17	2.651	3.000
118	Vietnam	VI	33	1.906	4.930
119	Yemen	YE	14	3.399	2.728
120	Zambia	ZM	38	1.783	4.917
121	Zimbabwe	ZW	22	2.322	3.760

	GDP Growth	Cluster Lables
0	3.403	2
1	4.202	2
2	-2.298	1
3	0.208	1
4	2.471	0
5	1.482	0
6	-3.772	1
7	2.915	2
8	6.923	2
9	1.239	0
10	4.027	2
11	4.100	2

12	2.500	2
13	2.864	0
14	-3.595	1
15	3.443	2
16	-1.042	1
17	4.443	2
18	1.433	0
19	-6.373	1
20	1.555	0
21	6.700	2
22	1.960	2
23	4.329	0
24	2.930	2
25	2.836	2
26	2.408	0
27	2.400	1
28	1.137	0
29	6.567	2
30	-2.168	1
31	4.298	2
32	2.400	2
33	1.573	0
34	7.958	2
35	1.387	0
36	1.213	0
37	2.263	2
38	2.723	2
39	1.774	0
40	4.041	2
41	0.012	2
42	3.000	2
43	3.600	2
44	2.000	0
45	7.200	0
46	6.828	2
47	5.016	2
48	6.544	2
49	5.222	0
50	4.003	2
51	0.880	0
52	1.540	2
53	0.999	0
54	2.100	2
55	1.083	2
56	6.004	2
57	2.828	0
58	2.458	2

59	1.954	2
60	1.000	1
61	-1.170	1
62	2.274	0
63	4.138	2
64	2.270	2
65	4.239	0
66	5.370	2
67	1.522	2
68	3.600	0
69	2.302	2
70	4.000	2
71	2.358	2
72	1.487	2
73	3.400	2
74	0.561	2
75	2.095	0
76	3.956	0
77	4.700	2
78	-1.541	1
79	1.024	0
80	3.057	2
81	4.707	2
82	5.000	2
83	4.093	2
84	3.897	2
85	6.839	2
86	2.830	0
87	1.432	0
88	2.683	0
89	4.785	2
90	-0.248	1
91	5.932	2
92	1.398	2
93	6.574	2
94	2.779	2
95	4.903	2
96	1.996	0
97	3.285	0
98	2.492	0
99	0.279	0
100	3.235	0
101	4.295	2
102	3.310	0
103	1.310	0
104	1.402	0
105	6.579	2

106	3.230	2
107	2.406	2
108	-5.110	1
109	1.000	2
110	2.876	2
111	4.673	2
112	2.308	1
113	2.732	0
114	1.806	0
115	1.616	0
116	1.438	0
117	-18.000	1
118	6.211	2
119	-9.779	1
120	2.967	2
121	0.522	1

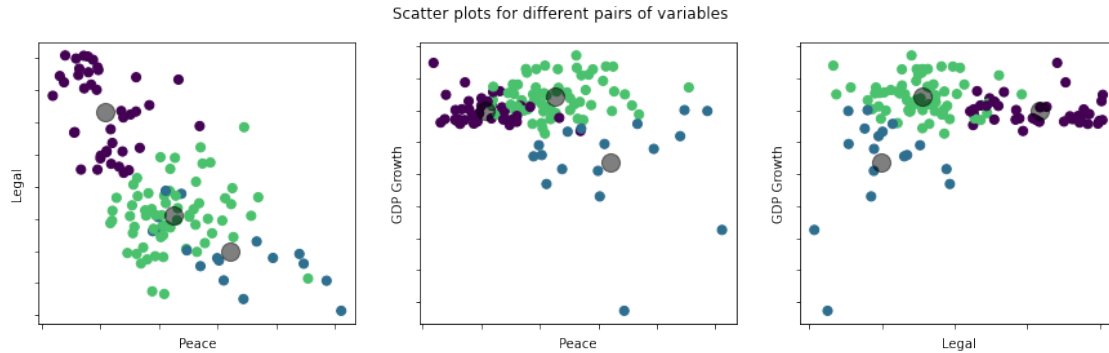
It appears that Cluster 1 contains the countries with lowest risk, Cluster 0 contains the countries with medium risk, and Cluster 2 contains the countries with highest risk

0.3.9 Visualize the result (2D plots)

```
[15]: # Scatter plots of different pairs of variables
fig, axs = plt.subplots(1,3,figsize=(15,4))
fig.suptitle('Scatter plots for different pairs of variables')
Labels = ['Peace','Legal','GDP Growth']
y = kmeans.labels_

# set up the color
norm = clrs.Normalize(vmin=0.,vmax=y.max()+0.8)
cmap = cm.viridis

# Plot the Figure
m = 0
for i in range(len(Labels)-1):
    for j in range(1,len(Labels)-i):
        axs.flat[m].scatter(X[Labels[i]], X[Labels[i+j]], c=cmap(norm(y)), s=50)
        axs.flat[m].scatter(Cluster_Table[Labels[i]],
↪Cluster_Table[Labels[i+j]], c='black', s=200, alpha=0.5)
        axs.flat[m].set_xlabel(Labels[i])
        axs.flat[m].set_ylabel(Labels[i+j])
        # Turn off tick labels.
        axs.flat[m].set_yticklabels([])
        axs.flat[m].set_xticklabels([])
    m = m+1
```



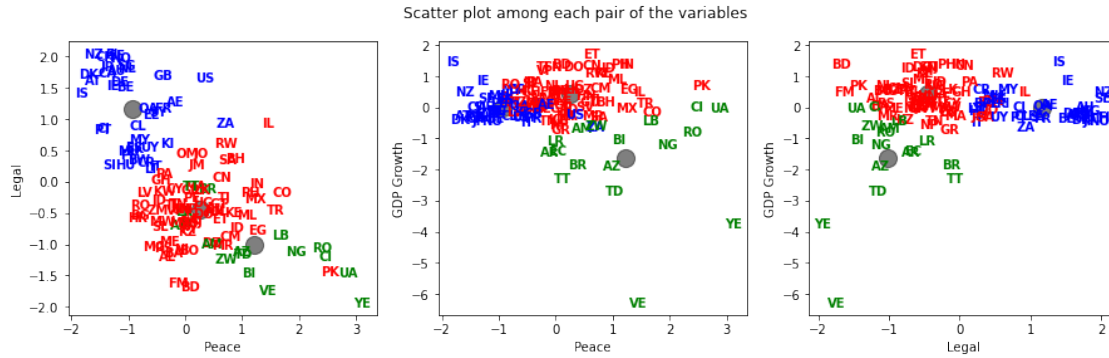
0.3.10 Visualize the result (2D plots with names of countries)

plot country abbreviations instead of dots.

```
[16]: # Plot of the Scatter Figure among each pair
fig, axs = plt.subplots(1,3,figsize=(15,4))
fig.suptitle('Scatter plot among each pair of the variables')
Labels=['Peace','Legal','GDP Growth']
colors = ['blue','green', 'red']

# set up the color
norm = clr.Normalized(vmin=0.,vmax=y.max()+0.8)
cmap = cm.viridis

# Plot the Figure
m = 0
for i in range(len(Labels)-1):
    for j in range(1,len(Labels)-i):
        axs.flat[m].scatter(X[Labels[i]],X[Labels[i+j]],c=y, s=0, alpha=0)
        axs.flat[m].scatter(Cluster_Table[Labels[i]],
        ↪Cluster_Table[Labels[i+j]], c='black', s=200, alpha=0.5)
        axs.flat[m].set_xlabel(Labels[i])
        axs.flat[m].set_ylabel(Labels[i+j])
        for h in range(X.shape[0]):
            axs.flat[m].text(X[Labels[i]].iloc[h], X[Labels[i+j]].iloc[h],
            ↪raw['Abbrev'].iloc[h],
                                color=colors[y[h]], weight='semibold', horizontalalignment =
            ↪'center', verticalalignment = 'center')
        m = m+1
```



0.3.11 Visualize the result (3D plot)

```
[17]: # set up the color
norm = clr.Normalizer(vmin=0.,vmax=y.max()+0.8)
cmap = cm.viridis

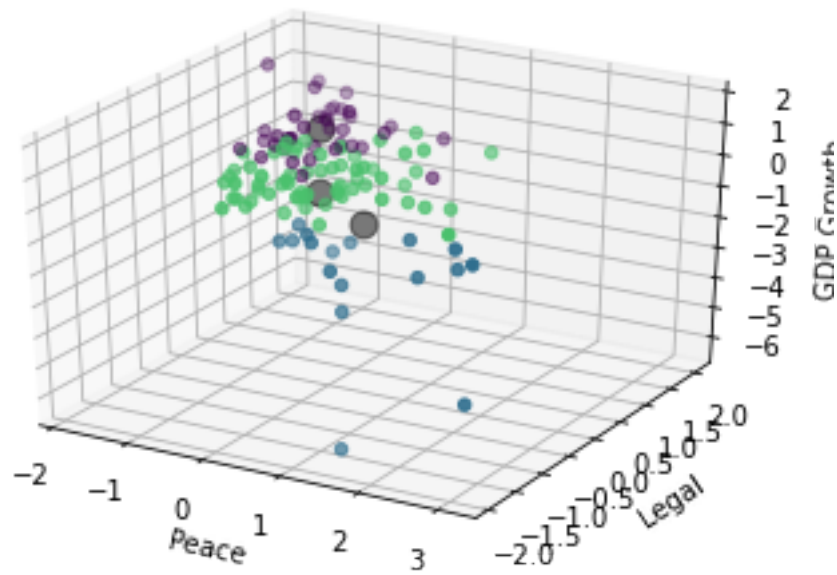
fig = plt.figure()
ax = fig.add_subplot(111, projection='3d')

ax.scatter(X.iloc[:,0], X.iloc[:,1], X.iloc[:,2], c=cmap(norm(y)), marker='o')

centers = kmeans.cluster_centers_
ax.scatter(centers[:, 0], centers[:, 1], c='black', s=100, alpha=0.5)

ax.set_xlabel('Peace')
ax.set_ylabel('Legal')
ax.set_zlabel('GDP Growth')

plt.show()
```



0.3.12 List the result

```
[18]: result = pd.DataFrame({'Country':raw['Country'], 'Abbrev':raw['Abbrev'],
    ↳ 'Label':y})
with pd.option_context('display.max_rows', None, 'display.max_columns', 3):
    print(result.sort_values(by=['Label', 'Country']))
```

	Country	Abbrev	Label
4	Australia	AU	0
5	Austria	AT	0
9	Belgium	BE	0
13	Botswana	BW	0
18	Canada	CA	0
20	Chile	CL	0
23	Costa Rica	CR	0
26	Czech Republic	CI	0
28	Denmark	DK	0
33	Estonia	EE	0
35	Finland	FI	0
36	France	FR	0
39	Germany	DE	0
44	Hungary	HU	0
45	Iceland	IS	0
49	Ireland	IE	0
51	Italy	IT	0
53	Japan	JP	0
57	Korea (South)	KI	0

62	Lithuania	LT	0
65	Malaysia	MY	0
68	Mauritius	MU	0
75	Netherlands	NL	0
76	New Zealand	NZ	0
79	Norway	NO	0
86	Poland	PL	0
87	Portugal	PT	0
88	Qatar	QA	0
96	Singapore	SG	0
97	Slovakia	SK	0
98	Slovenia	SI	0
99	South Africa	ZA	0
100	Spain	ES	0
102	Sweden	SE	0
103	Switzerland	CH	0
104	Taiwan	SY	0
113	United Arab Emirates	AE	0
114	United Kingdom	GB	0
115	United States	US	0
116	Uruguay	UY	0
2	Argentina	AR	1
3	Armenia	AM	1
6	Azerbaijan	AZ	1
14	Brazil	BR	1
16	Burundi	BI	1
19	Chad	TD	1
27	Democratic Republic of Congo	CI	1
30	Ecuador	EC	1
60	Lebanon	LB	1
61	Liberia	LR	1
78	Nigeria	NG	1
90	Russia	RO	1
108	Trinidad and Tobago	TT	1
112	Ukraine	UA	1
117	Venezuela	VE	1
119	Yemen	YE	1
121	Zimbabwe	ZW	1
0	Albania	AL	2
1	Algeria	DZ	2
7	Bahrain	BH	2
8	Bangladesh	BD	2
10	Benin	BJ	2
11	Bolivia	BO	2
12	Bosnia and Herzegovina	BA	2
15	Bulgaria	BG	2
17	Cameroon	CM	2
21	China	CN	2

22	Colombia	CO	2
24	Croatia	HR	2
25	Cyprus	CY	2
29	Dominican Republic	DO	2
31	Egypt	EG	2
32	El Salvador	SV	2
34	Ethiopia	ET	2
37	Gabon	GA	2
38	Georgia	GE	2
40	Ghana	GH	2
41	Greece	GR	2
42	Guatemala	GT	2
43	Honduras	HN	2
46	India	IN	2
47	Indonesia	ID	2
48	Iran	ID	2
50	Israel	IL	2
52	Jamaica	JM	2
54	Jordan	JO	2
55	Kazakhstan	KZ	2
56	Kenya	KE	2
58	Kuwait	KW	2
59	Latvia	LV	2
63	Madagascar	MG	2
64	Malawi	MW	2
66	Mali	ML	2
67	Mauritania	MR	2
69	Mexico	MX	2
70	Moldova	FM	2
71	Montenegro	ME	2
72	Morocco	MA	2
73	Mozambique	MZ	2
74	Nepal	NP	2
77	Nicaragua	NI	2
80	Oman	OM	2
81	Pakistan	PK	2
82	Panama	PA	2
83	Paraguay	PY	2
84	Peru	PE	2
85	Philippines	PH	2
89	Romania	RO	2
91	Rwanda	RW	2
92	Saudi Arabia	SA	2
93	Senegal	SN	2
94	Serbia	RS	2
95	Sierra Leone	SL	2
101	Sri Lanka	LK	2
105	Tanzania	TJ	2

106	Thailand	TJ	2
107	The FYR of Macedonia	TJ	2
109	Tunisia	TN	2
110	Turkey	TR	2
111	Uganda	UG	2
118	Vietnam	VI	2
120	Zambia	ZM	2